

Computer-Aided Two-Dimensional Analysis of Bipolar Transistors

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Abstract—A method for solving numerically the two-dimensional (2D) semiconductor steady-state transport equations is described. The principles of this method have been published earlier [1]. This paper discusses in detail the method and a number of considerable improvements. Poisson's equation and the two continuity equations are discretized on two networks of different rectangular meshes. The 2D continuity equations are approximated by a set of difference equations assuming that the hole and electron current density components along the meshlines are constant between two neighboring meshpoints in a way similar to that used by Gummel and Scharfetter [2] for the one-dimensional (1D) continuity equations. The resulting difference approximations have generally a much larger validity range than the conventional difference formulations where it is assumed that the change in electrostatic potential between two neighboring points is small compared with kT/q . Therefore, a much smaller number of meshpoints is necessary than for the conventional difference approximations. This reduces considerably the computation time and the required memory space. It will be shown that the matrix of the coefficients of this set of difference equations is always positive definite. This is an important property and guarantees convergence and stability of the numerical solution of the continuity equations.

The way in which the difference approximations for the continuity equations are derived gives directly consistent expressions for the current densities that can be used for calculating the currents. In order to demonstrate the kind of solutions obtainable, steady-state results for a bipolar n-p-n silicon transistor are presented and discussed.

I. INTRODUCTION

BY means of numerical solution methods it is possible to solve the general semiconductor transport equations accurately without the conventional restrictions such as locally neutral or space-charge regions, constant mobilities, simplified doping profiles, etc.

In 1964 Gummel [3] described a very attractive and efficient method for solving the 1D steady-state carrier transport equations. Later many other authors published solution methods not only for steady-state but also for small-signal and transient operation of 1D diode and transistor structures [4]–[15]. With these methods it is possible to study how physical parameters such as doping profile, carrier mobilities, lifetimes, and geometry are related to the electrical behavior of the device and to get a clear insight into high-level effects that are of growing importance for device optimization and design of accurate circuit models.

Lateral variations in current densities and potentials

and their relationship with base current cannot be taken into account in the 1D transistor structures. Quasi-2D models have been made to include these 2D effects [16], [17]. They consist of a number of 1D transistor models coupled with lateral base resistances. The basic assumption is that, apart from the base current, the transistor behavior is 1D. In fact, as will be shown by our rigorous 2D transistor calculations, this is not true when current spreading is present. Particularly for high current densities when base widening occurs (see, e.g., [18], [19]), current and voltage dependent current spreading takes place directly behind the emitter-base junction (see Section VII). It is also difficult to represent realistically the emitter sidewall injection in a quasi-2D model. Other approximate 2D transistor models have been made by splitting up the 2D transistor structure into neutral and space-charge regions [20], [21].

However, in order to analyze accurately transistors and other semiconductor structures under all kinds of operation conditions, the general basic carrier transport equations must be solved rigorously:

$$\nabla^2\psi = -q/\epsilon(p - n + N_D^+ - N_A^-) \quad (\text{Poisson's equation}) \quad (1)$$

$$\begin{aligned} J_p &= -q\mu_p p \nabla\psi - qD_p \nabla p \\ J_n &= -q\mu_n n \nabla\psi + qD_n \nabla n \end{aligned} \quad (\text{Current density equations}) \quad (2)$$

$$\begin{aligned} \nabla \cdot J_p + q\partial p/\partial t &= -qR \\ \nabla \cdot J_n - q\partial n/\partial t &= qR \end{aligned} \quad (\text{Continuity equations}). \quad (3)$$

Kennedy and O'Brien [22]–[24] have presented detailed graphical results of 2D steady-state calculations for junction field-effect transistors (JFET's). Unfortunately, they give no details of the way in which these equations are solved. Kim and Yang [25] consider in their numerical analysis of JFET's the majority carriers only and assume that the quasi-Fermi potential is one dimensional. This makes direct integration of the current density equation possible, which simplifies appreciably the numerical solution.

Recently it has been shown [1] how the 1D steady-state iterative scheme given by Gummel [3] can be extended to the 2D problem. This method will be described briefly. Normalizing (1), (2), and (3) in the same way as de Mari [4] did, the steady-state equations become

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$$\nabla^2 \psi = n - p - N \quad (4)$$

$$J_p = -\gamma_p^{-1}(p \nabla \psi + \nabla p)$$

$$J_n = -\gamma_n^{-1}(n \nabla \psi - \nabla n) \quad (5)$$

$$\nabla \cdot J_p = -R$$

$$\nabla \cdot J_n = R. \quad (6)$$

Here γ_p^{-1} and γ_n^{-1} are the normalized carrier mobilities. In the expressions for the current densities the Einstein relationship between the diffusion coefficient and the mobility is used.

With two new variables ϕ_p and ϕ_n , defined by

$$\begin{aligned} \phi_p &= \exp(\varphi_p) \\ \phi_n &= \exp(-\varphi_n) \end{aligned} \quad (7)$$

where φ_p and φ_n are the hole and electron quasi-Fermi potentials

$$\begin{aligned} \varphi_p &= \psi + \ln(p) \\ \varphi_n &= \psi - \ln(n) \end{aligned} \quad (8)$$

the basic equations (4)–(6) reduce to three coupled elliptic partial differential equations in ψ , ϕ_p , and ϕ_n :

$$\nabla^2 \psi = \phi_n \exp(\psi) - \phi_p \exp(-\psi) - N \quad (9)$$

$$\nabla \cdot \{\gamma_p^{-1} \exp(-\psi) \nabla \phi_p\} = R \quad (10)$$

$$\nabla \cdot \{\gamma_n^{-1} \exp(\psi) \nabla \phi_n\} = R \quad (11)$$

with

$$J_p = -\gamma_p^{-1} \exp(-\psi) \nabla \phi_p \quad (12a)$$

$$J_n = \gamma_n^{-1} \exp(\psi) \nabla \phi_n. \quad (12b)$$

These three equations are discretized on a nonuniform mesh. Poisson's equation, (9), is considered as a nonlinear equation in ψ and is linearized according to the Newton method (see Section III). The two continuity equations (10) and (11) are linear self-adjoint elliptic partial differential equations in ϕ_p and ϕ_n . The sequence of solution of this system of equations is the same as in the 1D Gummel method; first the nonlinear Poisson equation is solved, assuming ϕ_p and ϕ_n are known, and next each of the continuity equations is solved using the just calculated electric potentials from the Poisson equation. This cycle is iterated until a sufficient accuracy is reached (see Fig. 1). The main difference between the 1D and the 2D solution method is the way each of these equations is solved numerically. In the 2D analysis the equations are solved by iterative solution methods, such as the method of successive overrelaxation (SOR) or successive line overrelaxation (SLOR) [26]. In the 1D solution process, on the other hand, the linearized Poisson equation can be solved directly by Gaussian elimination and the two continuity equations by integration. This difference in solution method introduces severe complications for the 2D solution process. The coefficient matrices of the continuity difference equations depend upon the electric potential ψ , which changes with each "outer" iteration. Therefore, it is necessary that independent of the values of the elec-

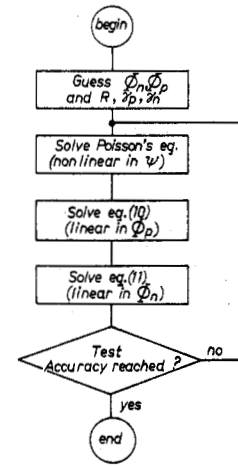


Fig. 1. Flow diagram of the iterative scheme.

tric potential in each meshpoint the difference equations are numerically stable and that the iterative solution method of the individual equations is always convergent. Though the difference approximations described in [1] satisfy these conditions, we will derive in this paper other difference approximations that are more accurate and have also the mentioned properties.

With this method, taking into account the oxide and the oxide-semiconductor interface, Heydemann [27] and van Dorpe and Xuong [28] have analyzed a metal-oxide-semiconductor (MOS) transistor. The last mentioned authors have refined the difference equations for the continuity equations in order to avoid computational difficulties for high applied voltages. Kilpatrick and Ryan [29] used the method for analyzing a lateral p-n-p transistor. Dubock [30] uses a special difference formulation for the continuity equations. Depending upon the sign of the electric field at each meshpoint a suitable difference approximation is chosen. The difference approximation thus changes in the different meshpoints, which seems unattractive from a numerical point of view. Reiser [31] has published a method for the solution of the time dependent 2D equations. Assuming that the space-charge term in (4) is known, this equation is solved by means of Hockney's method [32]. Although this is a very rapid and direct method of solution, it has some restrictions and cannot be used to solve the Newton-linearized Poisson equation. Therefore, the rapid rate of convergence of the Newton process is lost, which is especially important for low injection. Another restriction is that constant stepsizes in both directions are needed. This can introduce an excessive number of meshpoints in areas where they are not needed.

In a recent paper [33] van Dorpe, Borel, Merkel, and Saintot describe a 2D numerical analysis of the MOS transistor. The used difference approximations are the same as already given in [27]. At the end of their program they calculate the longitudinal component of the hole current density via an integral formulation because of the great inaccuracy met when

the current density is calculated from numerically differentiation. Heydemann uses in his last paper [34] the strongly implicit method of Stone [35] for solving each of the three systems of difference equations instead of the more usual SOR or SLOR methods.

An important drawback of the above-mentioned methods is the large number of meshpoints involved for accurate calculations. This is caused by the used difference approximations and the way of solving. Rather large computation time and required memory space are the consequence of it. Usually the number of iterations for each of the three equations depends rather strongly upon the number of meshpoints, which increases the computation time even more.

We will show that the efficient difference approximation, which Gummel and Scharfetter used for their analysis of a silicon Read diode oscillator [2], can be extended to two dimensions. Assuming that the hole and electron current density components along the meshlines are constant between two neighboring meshpoints, it is possible to derive difference approximations for (10) and (11) that have a much larger validity range than those used in the above-mentioned methods, which are based on the assumption that the change in electrostatic potential between two meshpoints is small compared to kT/q . The large validity range of the new difference approximations requires less meshpoints for accurate calculations.

We will prove that the matrix built up from the coefficients of the difference equations is always positive definite [26]. This property is very important for the numerical solution because it guarantees numerical stability and convergence of, e.g., SOR and SLOR.

The use of different meshes that are optimal for each of the three equations reduces the total number of meshpoints again.

II. DATA AND BOUNDARY CONDITIONS OF THE USED EXAMPLE

We shall illustrate our method with an example of a bipolar n-p-n silicon transistor. The geometrical structure is given in Fig. 2. The doping profile is calculated from the 2D diffusion equation [37]. The doping profile along the X axis and under the base contact is shown in Fig. 3. The generation-recombination term is given by the Hall, Shockley, and Read model [38]

$$R = (pn - n_i^2) / \{ \tau_{n0}(p + p_1) + \tau_{p0}(n + n_1) \} \quad (13)$$

where

- τ_{n0}, τ_{p0} electron and hole lifetime;
- p_1, n_1 the hole and electron concentration that would exist if the Fermi level were at the trap level.

We will assume that the recombination centers are near the center of the bandgap:

$$n_1 = p_1 = n_i.$$

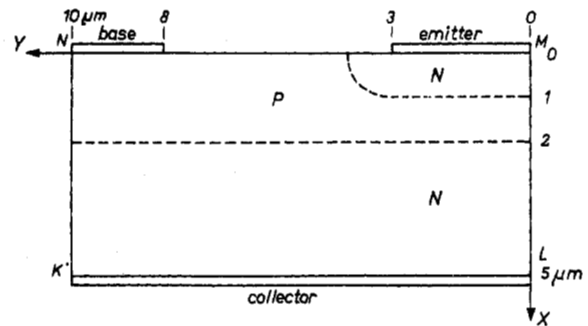


Fig. 2. Geometrical structure of n-p-n silicon transistor. Metallurgical junctions are indicated.

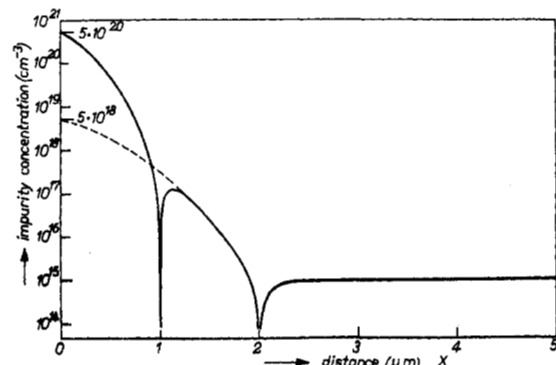


Fig. 3. Doping profile along ML and NK (see Fig. 2).

And further we have taken in our example

$$\tau_{p0} = \tau_{n0} = 0.5 \times 10^{-7} \text{ s.}$$

The mobility is a function of doping and electric field [3], [39]. The boundary conditions are the same as described in [1]. At the ohmic contacts the electrostatic and quasi-Fermi potentials are known from the assumed charge neutrality and equilibrium and from the given applied voltages. Along the other parts of the boundary the normal derivatives of the potentials are zero. For analyzing MOS transistors, more realistic boundary conditions at the interface with the oxide layer can be used [27], [28]. The calculations and computer plots were made on a rather small Philips ELX8 computer with a memory capacity of about 23K words of 27 b.

III. POISSON'S EQUATION

Poisson's equation, (9), is considered as a nonlinear equation in ψ while ϕ_p and ϕ_n are known. This equation is linearized in the usual way [1], [3]:

$$\nabla^2 \delta - a^2 \delta = b \quad (14)$$

where

- δ a perturbation on ψ_0 ;
- $a^2 = \phi_n \exp(\psi_0) + \phi_p \exp(-\psi_0) = n + p$;
- $b = -\nabla^2 \psi_0 + \phi_n \exp(\psi_0) - \phi_p \exp(-\psi_0) - N$
 $= -\nabla^2 \psi_0 + n - p - N$;
- ψ_0 the ψ values of the former iteration.

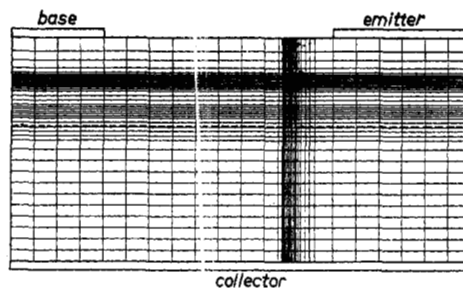


Fig. 4. Nonuniform mesh for discretizing Poisson's equation (E -net; 46×41 meshpoints).

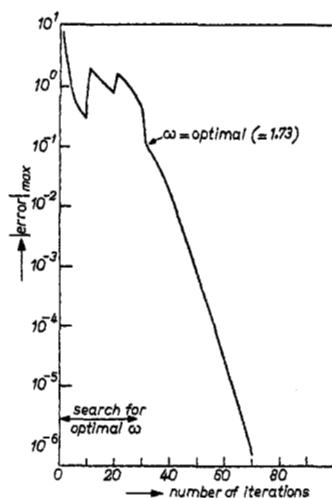


Fig. 5. Rate of convergence of the δ -equation solved by SOR with ω as overrelaxation parameter.

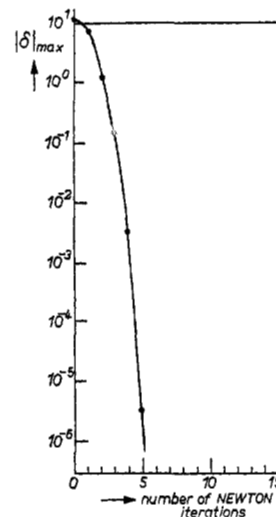


Fig. 6. Rate of convergence of the Newton process.

Because a^2 is a function of x and y (14) cannot be solved by Hockney's method as mentioned before. This linear elliptic partial differential equation is discretized on a nonuniform mesh (see Fig. 4) and replaced by a set of finite difference equations by means of the box integration method [40]. This net of meshpoints is called the E -net.

The problem can be written in matrix notation as

$$A\delta = b. \quad (15)$$

The matrix A is a symmetric and positive definite matrix with diagonal dominance. Therefore, the usual iterative solution methods are convergent [26]. Even with this very nonuniform mesh the rate of convergence is sufficient (see Fig. 5) because of the presence of the term $-a^2\delta$, which provides an extra diagonal dominance. The solution of the linearized Poisson equation is used to correct the ψ_0 values. This is one Newton iteration. By recalculating the coefficients a^2 and b , the process can be repeated, etc. Fig. 6 gives the quadratic rate of convergence of this Newton process. Usually it is sufficient to correct the ψ_0 values only once with the calculated δ values and then continue with the solution of the continuity equations.

IV. CONTINUITY EQUATION

Consider the continuity equation for the holes, and omit the subscript p for the sake of simplicity; the corresponding equation for the electrons can be treated in the same way. The functions γ , ψ , and R are assumed to be known in the meshpoints of the E -net.

In Appendix A it is shown in detail how the Gummel-Scharfetter difference approximation for the current density, formulated in the variable ϕ , is extended to two dimensions. The two continuity equations are discretized on a second network of meshpoints, the so-called J -net (see Fig. 7). It is assumed that the current density component along the meshlines is constant between two neighboring meshpoints of the J -net, while the electric field component along the meshlines is assumed constant between two neighboring meshpoints of the E -net. The conventionally used difference approximations are only valid when the change in electric potential between two meshpoints is less than kT/q . In a reverse-biased base-collector junction this would require very small stepsizes [41].

According to these new approximations we use two different networks of meshpoints, the already mentioned E -net and J -net. The resulting system of dif-

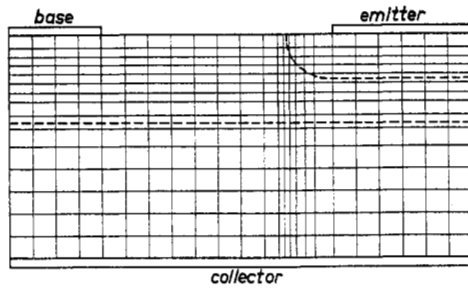


Fig. 7. Nonuniform mesh for discretizing the continuity equation (J -net; 16×25 meshpoints).

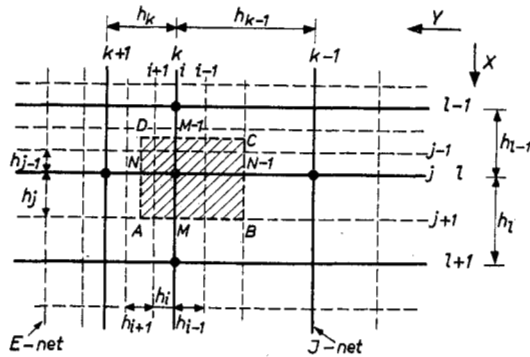


Fig. 8. The E -net is a refinement of the J -net. Meshpoints of the E -net are indicated by the integers j, i and of the J -net by l, k . Variable step sizes in the x and y -directions are indicated for both nets. Points $M, M-1, N$, and $N-1$ are midway between two neighboring meshpoints.

ference equations can be written in matrix notation as (see Appendix A)

$$A\phi = b. \quad (16)$$

The coefficient matrix A is symmetric and positive definite. This makes it possible to solve this problem by standard numerical techniques [26]. Because the J -net needs generally a much smaller number of meshpoints, which are more uniformly distributed than the meshpoints of the E -net, the rate of convergence of SOR and SLOR is much improved.

The "positive" properties of the matrix A are also important for the difference approximations of the time dependent continuity equation (see Appendix C).

From the quasi-Fermi potentials in the corners of each cell of the J -net (points A, B, C , and D in Fig. 8) we derive by means of bilinear interpolation the quasi-Fermi potentials in the meshpoints of the E -net. These are needed for the solution of Poisson's equation and for the calculation of the recombination in every meshpoint of the E -net.

In order to avoid very large values for ϕ_p and ϕ_n these variables are scaled down by means of a similarity transformation (see Appendixes A and B). Then a system of difference equations in the variable p results [see Appendix A, (A12)]:

$$Tp = b. \quad (17)$$

T is the transformed coefficient matrix A .

All the quantities in these difference equations are bounded in value and no problems with overflow can occur. The eigenvalue spectrum of the original coefficient matrix is not changed when a similarity transformation is applied. Consequently, the iterative solution process (e.g., SOR and SLOR) behaves in the same way for these difference equations.

V. CURRENT, CURRENT DENSITY, AND CUTOFF FREQUENCY

At every midpoint between two meshpoints of the J -net the current density can be calculated from the same difference approximation for the current density as is used for the continuity equation:

$$J_M = -a_M \{ p_{l+1,k} \exp(\psi_{l+1,k} - \psi_{l,k}) - p_{l,k} \}. \quad (18)$$

This follows from transformation of (A9) (see Appendix A) to the variable p . When the quasi-Fermi potential is locally nearly constant this calculation introduces numerical inaccuracy because nearly equal numbers are subtracted. Therefore, it is difficult in this way to calculate the electron current density in the quasi-neutral emitter region or the hole current density underneath the base contact. Apart from the difficulties in these areas the total current flowing through a line $x = \text{constant}$ (see Fig. 2) is easily calculated by integrating the current density along the line. The collector current is calculated in this way by integrating along a line $x = \text{constant}$ just before the collector contact. The base current is given by

$$I_B = - \iint_{\text{total transistor area}} R \, dx \, dy. \quad (19)$$

This follows from integration of the hole continuity equation on the assumptions that the base current consists of holes only and that holes are fed into the transistor via the base contact only.

From I_C and I_B the current amplification factor is calculated:

$$h_{FE} = I_C / I_B. \quad (20)$$

The cutoff frequency is calculated in accordance with the charge control principle from a perturbation of the charge concentration and the resulting change in collector current:

$$f_T = (1/2\pi) \Delta I_C / \Delta Q_p \quad (21)$$

where ΔQ_p is the change in total stored hole charge.

VI. CONSIDERATIONS ABOUT CONVERGENCE AND CHOICE OF MESHPOINTS

In all our computer calculations the "outer" iteration (see Fig. 1) converged well except for very long device

structures (e.g., larger than $100\ \mu\text{m}$) with heavy recombination in the whole device; the rate of convergence then can be extremely small which makes the method rather impractical under these conditions. This experience agrees with a recent study of this convergence problem [42].

In order to check the convergence behavior we test during each "outer" iteration the relative and absolute change in electric potential in each meshpoint. Also, the relative change in the carrier concentrations is checked in each meshpoint. In order to get an idea of the rate of convergence, we show in Fig. 9 the maximum absolute change in electric potential for two successive iterations as a function of the number of iterations. For low current levels the rate of convergence of the Newton-linearized Poisson equation dominates the overall convergence behavior and a nearly quadratic convergence results [compare the quadratic rate of convergence when Poisson's equation alone is solved by means of the Newton method (see Fig. 6)]. For higher current levels the mutual coupling between the equations becomes stronger and the convergence rate slows down (see Fig. 9). This rate of convergence can still be speeded up [43]. The convergence behavior is very similar to the 1D Gummel method.

The total current flowing through lines $x = \text{constant}$ is, as a function of x , constant within 1 percent. This conservation of current is self-evident because the expression for calculating the current density is also used in the difference approximation for the continuity equations.

Because one wants to use a minimum number of meshpoints it is very important to choose carefully an optimal distribution of meshpoints. This depends also upon the specific phenomena and characteristics of the transistor to be studied. We have found that often errors, introduced because of locally too few meshpoints, are kept very localized and hardly influence phenomena somewhere else or the overall characteristics.

We refined the J -net from 400 to 729 meshpoints and found only small differences (less than 5 percent) in the carrier concentrations and the electric potential. But when we reduced the number of meshpoints to only 247, differences in these variables of the order of 20 percent and in some few points of 100 percent took place. However, the calculated total currents in the $I_C V_{EB}$ characteristic (see Fig. 10) deviate for these three J -nets less than 1 percent. The total base current in the last network of 247 meshpoints deviates in about 20 percent of the values calculated with the finer meshes, which were nearly equal. The number of meshpoints of the E -net was the same for these three J -nets, namely 1886.

VII. DISCUSSION OF THE RESULTS

In a number of computer plots the internal as well as the external behavior of the n-p-n silicon transistor under normal and saturation conditions will be illustrated.

In Fig. 11 the electron current spreading under nor-

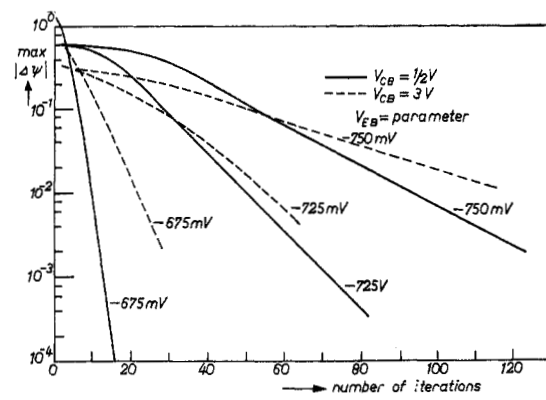


Fig. 9. Rate of convergence of the total system for different biases.

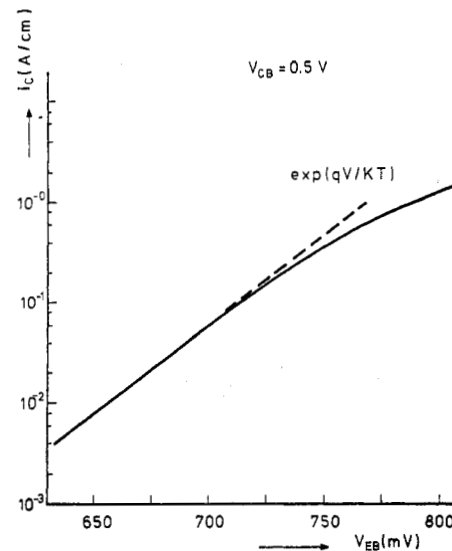


Fig. 10. Calculated $I_C V_{EB}$ characteristic for $V_{CB} = 0.5\ \text{V}$.

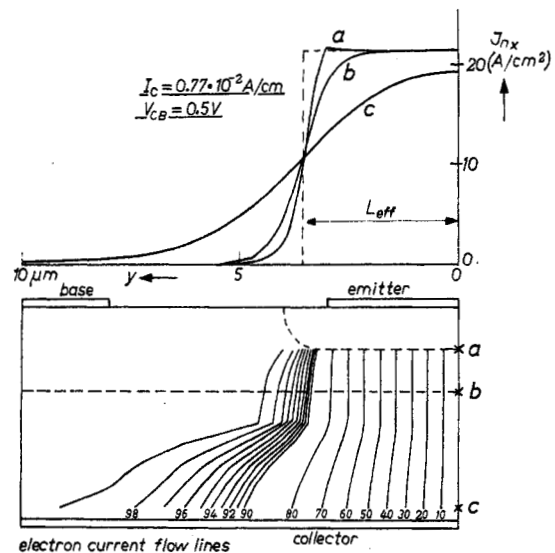


Fig. 11. Calculated electron current flow lines and current density distribution along the emitter junction (curve a), collector junction (curve b), and along the collector contact (curve c) are shown. The number under the flow line gives the percentage of the total current that flows between this line and the axis of symmetry (x axis).

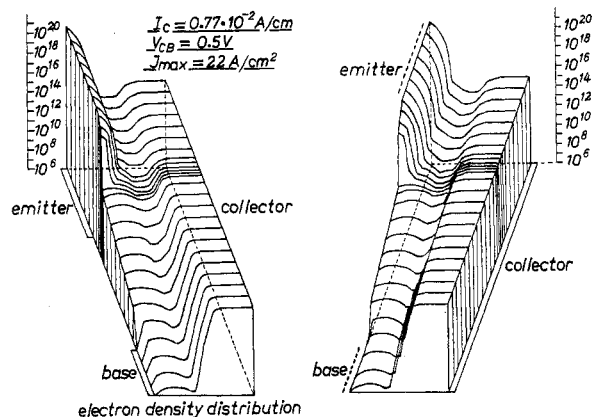


Fig. 12. Three-dimensionally represented electron density distributions seen from different angles. The concentration (cm^{-3}) is indicated along the vertical axis.

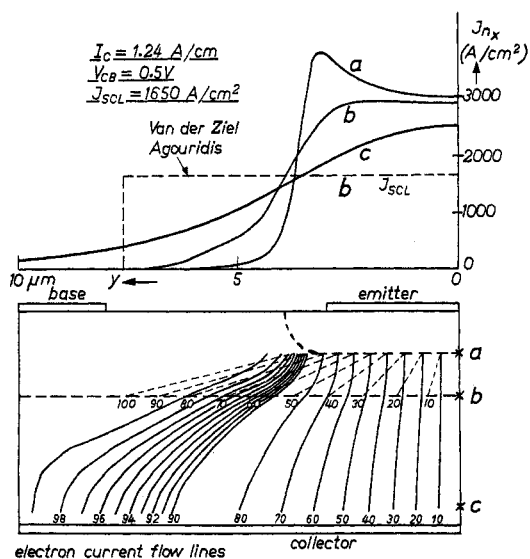


Fig. 13. Calculated electron current flow lines and current density distribution along the emitter junction (curve *a*), collector junction (curve *b*), and along the collector contact (curve *c*) are shown. The flow lines are labeled with the percentage of the total current flowing between this line and the x axis. The corresponding results of the model of van der Ziel and Agouridis [44] are indicated by dashed lines. Curve *a* shows the current crowding along the emitter-base junction.

mal operation is shown. Current spreading is only present in the quasi-neutral collector region. Three-dimensional pictures of the electron density distribution in the transistor under low injection are shown in Fig. 12. When the current is increased, the space-charge concentration in the base-collector junction decreases and finally disappears. Spreading of electron flow lines takes place more and more right behind the emitter-base junction dependent upon current and voltage (see Fig. 13). Consequently the x component of the electron current density is not constant in the x direction as it is in a 1D model (apart from the influence of recombination) but deviates appreciably from that value (see Fig. 14). For low current densities the x component of the electron current density along the line $y=0$ is nearly constant up to the point where the electrons

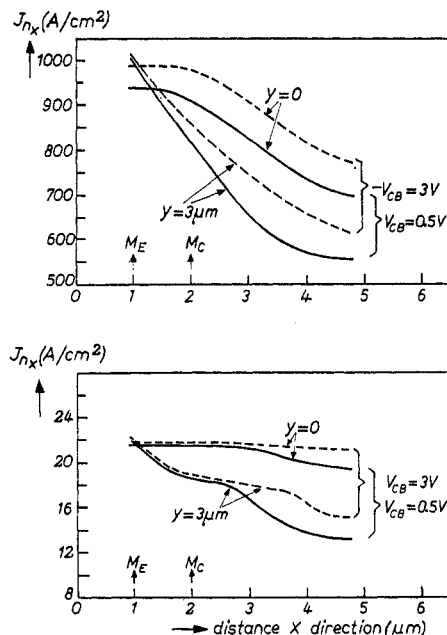


Fig. 14. The variation of the x component of the electron current density as a function of x is shown calculated along two lines, $y=0 \mu\text{m}$ and $y=3 \mu\text{m}$ (see Fig. 2). In the 1D model J_{n_x} would be constant (when recombination is neglected). The metallurgical emitter and collector junctions are indicated by M_E and M_C .

leave the collector space-charge layer and spread into the ohmic collector region. The current density then decreases because of the current spreading. When $V_{CB} = 3 \text{ V}$, very little spreading is possible because the ohmic collector region is very thin. At higher current densities we see from this figure that the current density already decreases behind the emitter-base junction. In this situation the electric field at the collector junction is very small and cannot focus the electrons any longer. The lines of equipotential and of constant hole concentration in Figs. 15 and 16 show how the well-known Kirk effect [18] works out in the lateral dimension.

In Fig. 17 it is shown how strongly the hole and electron concentrations vary along the base-collector junction in the lateral direction under several injection levels.

These rather strong lateral variations under the emitter area, together with the above-shown current and voltage dependent current spreading when the transistor is in saturation, are a severe difficulty in trying to model the transistor with a set of coupled 1D models.

In Fig. 13 current crowding along the emitter-base junction is shown. This crowding under saturation conditions is an effect that has not yet been explored. More numerical calculations are needed to be able to model crowding under these very complicated and nonlinear situations. In this figure we have also indicated how the current spreading would be according to the hypothesis of van der Ziel and Agouridis [44]. Their hypothesis states that there is a maximum current density, the so-called space-charge-limited current density J_{SCL} , when the current increases. This J_{SCL} is defined as that current density at which the electric

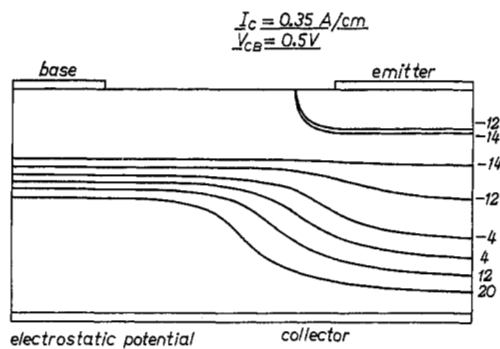


Fig. 15. Calculated lines of constant electrostatic potential. The niveau lines are labeled with the value of the potential normalized in kT/q units.

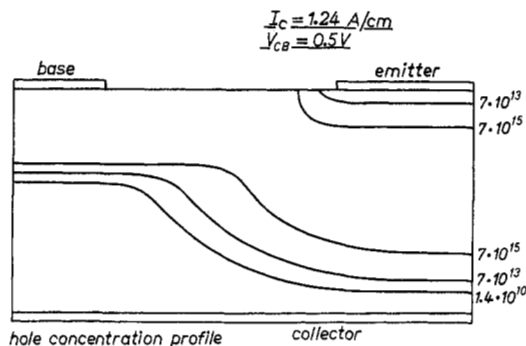


Fig. 16. Calculated lines of constant hole concentration. The niveau lines are labeled with the value of the hole concentration (cm^{-3}). The shape of the area where the hole charge is stored in the collector is clearly shown.

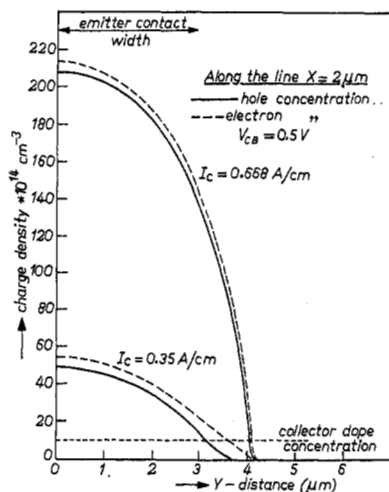


Fig. 17. Calculated hole and electron concentration along the base-collector junction ($x=2 \mu\text{m}$; see Fig. 2) as a function of y . The width of the emitter contact is indicated.

field at the base-collector junction has become zero. This current density is given by [45]

$$J_{SCL} = qv_{lim} \left(N_D + 2\epsilon/q \frac{|V_{CB} + V_0|}{W_C^2} \right). \quad (22)$$

"Any further increase in collector current must result in a widening of the region in which collector current flows" [44]. Our 2D calculations, as represented in

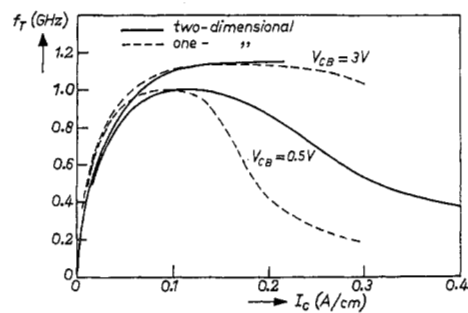


Fig. 18. 1D and 2D calculated cutoff frequency as a function of the current.

Fig. 13, show clearly that the current density is not limited by J_{SCL} . The current spreading in the base is much less than predicted by these authors, which is a result of their unrealistic assumption. More 2D numerical calculations are needed to construct a well-founded model for the current spreading under saturation conditions. In Fig. 18 the calculated cutoff frequency is given as a function of the current. The points of these characteristics that correspond with the presented and discussed plots of internal current density, potential, and carrier density distributions can easily be found. We have also indicated the corresponding results for the 1D transistor structure (that is, the transistor along the line $y=0$ in Fig. 2). In order to compare the 1D calculations with the 2D results, the current density in the 1D structure is multiplied by an effective emitter width L_{eff} . This follows from the consideration that for low injection the integral of the 2D calculated current density along the emitter-base junction must be equal to the product of L_{eff} and the 1D calculated current density (see Fig. 11). For low currents the f_T values are for the 2D structure somewhat smaller than in the 1D structure because of the extra emitter sidewall capacitance. For high current densities the f_T shows a later and slower falloff in the 2D structure than in the 1D structure, due to the current spreading in the 2D transistor. This effect of current spreading on the f_T characteristic is also shown by Slatter [47] in his cylindrical-geometry bipolar transistor structure.

VIII. CONCLUSION

An efficient and general numerical method for solving the carrier transport equations for 2D semiconductor device structures is presented and demonstrated for a bipolar n-p-n silicon transistor. The described difference approximations for the continuity equations allow rather coarse meshes while still being accurate. Because the coefficient matrices of the difference equations for the three basic partial differential equations (9), (10), and (11) are always positive definite, convergence and stability of the numerical solution of these three equations is guaranteed. The overall convergence was present in all our calculations and behaves very similar to the 1D Gummel method.

With the described method it is possible to analyze

important nonlinear effects such as: emitter-base crowding under extreme conditions as high injection in the base and saturation of the base-collector junction; lateral current spreading and charge storage; and how these effects are related to the external electrical behavior.

APPENDIX A

DIFFERENCE EQUATIONS FOR THE CONTINUITY EQUATIONS

We will consider the continuity equation for the holes and omit the subscript p . The normalized hole continuity equation is

$$\nabla \cdot J = -R, \quad \text{with } J = -\gamma^{-1} \exp(-\psi) \nabla \phi. \quad (\text{A1})$$

The continuity equation is discretized on the so-called J -net. It is assumed that the current density changes slowly per cell of the J -net, while the electric field varies slowly per cell of the E -net. For simplicity we assume that the E -net covers the J -net (see Fig. 8).

Using the box integration method [40] and the Gaussian theorem it follows that

$$\begin{aligned} \iint_{ABCD} \nabla \cdot J \, dx \, dy &= \oint J_{\text{normal}} \, ds \\ &= - \iint_{ABCD} R \, dx \, dy. \end{aligned} \quad (\text{A2})$$

Approximation of the integrals gives

$$\begin{aligned} \frac{1}{2}(h_k + h_{k-1})(J_M - J_{M-1}) + \frac{1}{2}(h_l + h_{l-1})(J_N - J_{N-1}) \\ = -\frac{1}{4}(h_k + h_{k-1})(h_l + h_{l-1})R_{\text{eff},l,k} \end{aligned} \quad (\text{A3})$$

where $R_{\text{eff},l,k}$ is the effective recombination in the rectangle $ABCD$ defined by

$$R_{\text{eff},l,k} = \frac{\sum_{j,i} R_{j,i} h_j h_i}{\frac{1}{4}(h_k + h_{k-1})(h_l + h_{l-1})}. \quad (\text{A4})$$

$R_{j,i}$ is the mean value of the recombination in the four corners of the cell (j, i) (see Fig. 8). Only that part of each cell (j, i) contributes to the summation in the numerator in as far as it lies within the rectangle $ABCD$. J_M , J_{M-1} , J_N , and J_{N-1} are the current density components along the meshlines at points midway between two meshpoints (see Fig. 8) and are assumed to be constant between the two meshpoints. Considering the interval $x_l \leq x \leq x_{l+1}$ along the line $y = y_k$ it follows that

$$J_x = -\gamma^{-1} \exp(-\psi) d\phi/dx = \text{const} (= J_M). \quad (\text{A5})$$

Instead of approximating this expression by numerical differentiation we first integrate and then approximate the integral. Then it follows that

$$J_M = - \frac{\phi_{l+1} - \phi_l}{\int_{x_l}^{x_{l+1}} \gamma(s) \exp\{\psi(s)\} ds}. \quad (\text{A6})$$

Define a coefficient

$$a_M = \frac{1}{\int_{x_l}^{x_{l+1}} \gamma(s) \exp\{\psi(s) - \psi_{l,k}\} ds}. \quad (\text{A7})$$

Assuming that the electric field between two meshpoints of the E -net is constant, we approximate this coefficient as follows:

$$\begin{aligned} a_M &\approx \frac{1}{\sum_j \int_{x_j}^{x_{j+1}} \gamma(s) \exp\{\psi(s) - \psi_{l,k}\} ds} \\ &= \frac{1}{\sum_j \gamma_{j+\frac{1}{2},i} h_j \exp(\psi_{j+\frac{1}{2},i} - \psi_{l,k}) sh(\Delta_{j+\frac{1}{2},i})} \end{aligned} \quad (\text{A8})$$

where

$$\begin{aligned} sh(x) &= \sinh(x)/x \\ \Delta_{j+\frac{1}{2},i} &= \frac{1}{2}(\psi_{j+1,i} - \psi_{j,i}) \\ \psi_{j+\frac{1}{2},i} &= \frac{1}{2}(\psi_{j+1,i} + \psi_{j,i}). \end{aligned}$$

From (A6) and (A8) it follows that

$$J_M = -a_M \exp(-\psi_{l,k})(\phi_{l+1,k} - \phi_{l,k}), \quad \text{with } a_M > 0. \quad (\text{A9})$$

Notice that the integral in the denominator of a_M is bounded in magnitude also when the electrostatic potential takes on large values, because we have $\exp(\psi - \psi_{l,k})$ instead of $\exp(\psi)$. In the same way, difference approximations can be derived for J_{M-1} , J_N and J_{N-1} . Substitution of these approximations in (A3) gives

$$\begin{aligned} (a_M + a_{M-1} + r a_N + r a_{N-1}) \exp(-\psi_{l,k}) \phi_{l,k} \\ - a_M \exp(-\psi_{l,k}) \phi_{l+1,k} - a_{M-1} \exp(-\psi_{l,k}) \phi_{l-1,k} \\ - r a_N \exp(-\psi_{l,k}) \phi_{l,k+1} - r a_{N-1} \exp(-\psi_{l,k}) \phi_{l,k-1} \\ = -\frac{1}{2}(h_l + h_{l-1}) R_{\text{eff},l,k} \end{aligned} \quad (\text{A10})$$

with

$$r = (h_l + h_{l-1})/(h_k + h_{k-1}).$$

In matrix notation the set of difference equations can be written as

$$A\phi = b. \quad (\text{A11})$$

Note that when the E -net and J -net coincide and if the stepsizes are small enough, this difference approximation is identical to the more conventional one [1]. The coefficient matrix A is symmetric with positive diagonal entries and nonpositive off-diagonal entries. Further, this matrix is diagonally dominant and positive definite. This system of difference equations is therefore always stable, independent of the ψ values during the iteration process and can be solved by standard numerical techniques [26].

Because the potentials ϕ_p and ϕ_n can take on large values, practical difficulties can arise, such as overflow of computer capacity [4], [28]. This can easily be avoided by applying a similarity transformation (see Appendix B) to the set of difference equations (A10)

$$\begin{aligned}
& (a_M + a_{M-1} + ra_N + ra_{N-1})p_{l,k} - a_M \exp(\psi_{l+1,k} \\
& - \psi_{l,k})p_{l+1,k} - a_{M-1} \exp(\psi_{l-1,k} - \psi_{l,k})p_{l-1,k} \\
& - ra_N \exp(\psi_{l,k+1} - \psi_{l,k})p_{l,k+1} - ra_{N-1} \exp(\psi_{l,k-1} \\
& - \psi_{l,k})p_{l,k-1} = -\frac{1}{2}(h_l + h_{l-1})R_{\text{eff},l,k}. \quad (\text{A12})
\end{aligned}$$

This final difference equation could have been derived more directly if we would have started with the variable p instead of ϕ_p . But then, however, it would not be easy to conclude from the resulting coefficient matrix, which then is not symmetric and not diagonally dominant, that the iterative solution methods are convergent. By the derivation as we have shown via the variable ϕ we have proven that this coefficient matrix is similar to a positive definite matrix.

APPENDIX B

SCALING BY MEANS OF A SIMILARITY TRANSFORMATION

The matrix B is said to be similar to the matrix A if a nonsingular matrix C exists such that $B = C^{-1}AC$. Matrix B is said to be obtained from matrix A by a similarity transformation [46]. In Appendix A we have shown that it is possible to find a difference approximation that results in a diagonally dominant and symmetric positive definite matrix A :

$$A\phi = b, \quad \text{with } A = (a_{i,j}). \quad (\text{B1})$$

When C is defined by the following diagonal matrix

$$C = \begin{bmatrix} & & & & & & 0 \\ & & & & & & 0 \\ & & & & & & \\ & & & & & & \\ & & & & \alpha_{i-1} & & \\ & & & & & \alpha_i & \\ & & & & & & \alpha_{i+1} \\ & & & & & & \\ & & & & & & \\ 0 & & & & & & \end{bmatrix} \quad (\text{B2})$$

(B1) can be transformed:

$$(CAC^{-1})(C\phi) = (Cb). \quad (\text{B3})$$

Working out these matrix multiplications we have at point i

$$\begin{aligned}
& (\alpha_i/\alpha_{i-1})a_{i,i-1}(\alpha_{i-1}\phi_{i-1}) + a_{i,i}(\alpha_i\phi_i) \\
& + (\alpha_i/\alpha_{i+1})a_{i,i+1}(\alpha_{i+1}\phi_{i+1}) = \alpha_i b_i. \quad (\text{B4})
\end{aligned}$$

The new matrix (CAC^{-1}) is generally not symmetric and diagonally dominant. However, the eigenvalue spectrum is invariant for a similarity transformation. The variable ϕ_i is transformed to a new variable $(\alpha_i\phi_i)$. The scaling factors α_i can be chosen in such a way that the new variable $(C\phi)$ varies smoothly. When we use $\exp(-\psi_i)$ as scaling factors we have a difference formu-

lation in the variable p . Because the eigenvalue spectrum of a matrix is invariant for a similarity transformation, the coefficient matrix of the difference equations in the variable p behaves in the same way as the matrix of the difference equations in the variable ϕ .

APPENDIX C

THE TIME-DEPENDENT CONTINUITY EQUATIONS

The time-dependent continuity equation for the holes is considered and the subscript p is skipped for simplicity. The corresponding equation for the electrons can be treated in the same way. The time-dependent continuity equation is

$$\nabla \cdot J + \partial/\partial t\{\phi \exp(-\psi)\} = -R. \quad (\text{C1})$$

Using the method of Crank-Nicolson we obtain the following difference equation for the unknown ϕ values at the time t_{n+1} :

$$\begin{aligned}
& \frac{1}{2}[A^{(n+1)}\phi^{(n+1)} + A^{(n)}\phi^{(n)}] + h/\Delta t[\phi^{(n+1)} \exp(-\psi^{(n+1)}) \\
& - \phi^{(n)} \exp(-\psi^{(n)})] = -^2r \quad (\text{C2})
\end{aligned}$$

or

$$\begin{aligned}
& [A^{(n+1)} + (2h/\Delta t) \exp(-\psi^{(n+1)})I]\phi^{(n+1)} \\
& = -[A^{(n)} - (2h/\Delta t) \exp(-\psi^{(n)})I]\phi^{(n)} - ^2r. \quad (\text{C3})
\end{aligned}$$

Equation (C3) is of the type $A^*\phi^{(n+1)} = b$. I is the unit matrix. The matrix A is the same as in the steady-state problem (see (A11)) with elements following from (A8) and (A9). The new coefficient matrix A^* consists of the already known matrix A with extra positive diagonal terms added. This extra diagonal dominance results in a larger rate of convergence. Because this matrix is also positive definite the usual overrelaxation methods can be used. The difference approximations used by Reiser [31] lead to a coefficient matrix with complex eigenvalues, and underrelaxation instead of overrelaxation has to be used, which usually results in a smaller rate of convergence [40].

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